

TASK 4 REPORT – EXPLOITATION & SYSTEM SECURITY

Title:

Penetration Testing and System Security Using Metasploitable2

Objective:

To learn the penetration testing workflow including reconnaissance, scanning, exploitation, password attacks, and system hardening in a controlled environment.

Tools Used:

- Kali Linux
- Metasploitable2
- Nmap
- Metasploit Framework
- Hydra
- John the Ripper
- UFW Firewall

Methodology:

1. Configured Kali Linux and Metasploitable2 in VirtualBox.
2. Verified connectivity between attacker and target systems.
3. Performed scanning and service detection.
4. Attempted exploitation using Metasploit.
5. Demonstrated password attacks.
6. Applied firewall hardening.

Activities Performed:

1. Reconnaissance and Scanning

Used ping and Nmap to identify active host and open services.

2. Exploitation with Metasploit

Loaded exploit modules and configured target settings.

3. Password Attacks

Used Hydra for brute-force login testing.

4. Hash Cracking

Used John the Ripper to analyze password hashes.

5. System Hardening

Enabled firewall and configured deny rules.

Findings:

- Target host was reachable.
- Services were identified successfully.
- Security tools executed correctly.
- Firewall protection rules were applied.

Mitigation:

- Keep systems patched.
- Use strong passwords.
- Disable unused services.
- Restrict network access with firewall.
- Monitor logs regularly.

Conclusion:

This task provided hands-on experience in penetration testing methodology and defensive hardening using industry tools.

Result:

Scanning, exploitation attempts, password testing, and system hardening were demonstrated successfully.